

Instructions to the Students:

1. All the questions are compulsory.
2. Use of non-programmable scientific calculators is allowed.
3. Assume suitable data wherever necessary and mention it clearly.

	Marks
Q. 1 Solve Any Two of the following.	12
A) What is Blockchain? Explain key elements of it.	6
1) Difference between public vs. private Blockchain.	6
C) Explain Merkel Tree and Merkel Tree Structure.	6
Q.2 Solve Any Two of the following.	12
A) What is Bitcoin Mining and How Does Bitcoin Mining Work?	6
B) Why Proof of Burn required and how does algorithm work?	6
C) How do P2P networks work? Explain its type.	6
Q. 3 Solve Any Two of the following.	12
A) Explain design issues for permissioned Blockchain with any 2 issues.	6
B) Explain Paxos algorithm in detail.	6
What is Byzantine general problem and How Bitcoin Solves the Byzantine General's Problem?	6
Q.4 Solve Any Two of the following.	12
A) How KYC on Blockchain can help?	6
B) How Blockchain benefits food security in future?	6
C) How does Digital Identity work today using Blockchain?	6
Q. 5 Solve Any Two of the following.	12
A) Explain Hyperledger Fabric and how does it work?	6
B) What are the benefits of Hyperledger Fabric?	6
C) Explain writing smart contract using Ethereum ?	6

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2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
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	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What are Cryptographic Hash Functions? And explain the properties of the hash function.	Understand	6
B) What are the key properties and features of smart contracts?	Understand	6
C) Define Blockchain technology and explain the types of Blockchain Technology.	Understand	6
Q.2 Solve Any Two of the following.		12
A) Write the difference between Proof of Work (PoW) and Proof of Stake (PoS) in blockchain.	Understand	6
B) How does a Bitcoin transaction progress from creation to confirmation in the Bitcoin network?	Analysis	6
C) What is Double Spending in Blockchain? Explain with an example.	Apply	6
Q. 3 Solve Any Two of the following.		12
A) What challenges arise in designing a permissioned blockchain?	Apply	6
B) What are the key safety rules in the Raft consensus protocol that ensure consistency and integrity in the replicated log of nodes?	Understand	6
C) What are the differences between permissioned and permissionless blockchains?	Analysis	6
Q.4 Solve Any Two of the following.		12
A) What potential benefits can the mortgage industry gain from adopting blockchain technology?	Apply	6
B) Give any examples of uses for blockchain technology, along with a justification.	Apply	6
C) What is KYC? Explain the Shared KYC model based on Blockchain.	Understand	6
Q. 5 Solve Any Two of the following.		12

A) What are the key components of the Hyperledger Fabric architecture, and how do they contribute to the functionality and security of enterprise-grade blockchain networks?	Understand	6
B) What is the difference between Ripple and Corda?	Analysis	6
C) What are the key differences between Hyperledger Fabric and Ethereum?	Analysis	6

*** All the Best ***